MANIDIPINE HYDROCHLORIDE

1 | WHAT IS MANIDIPINE?

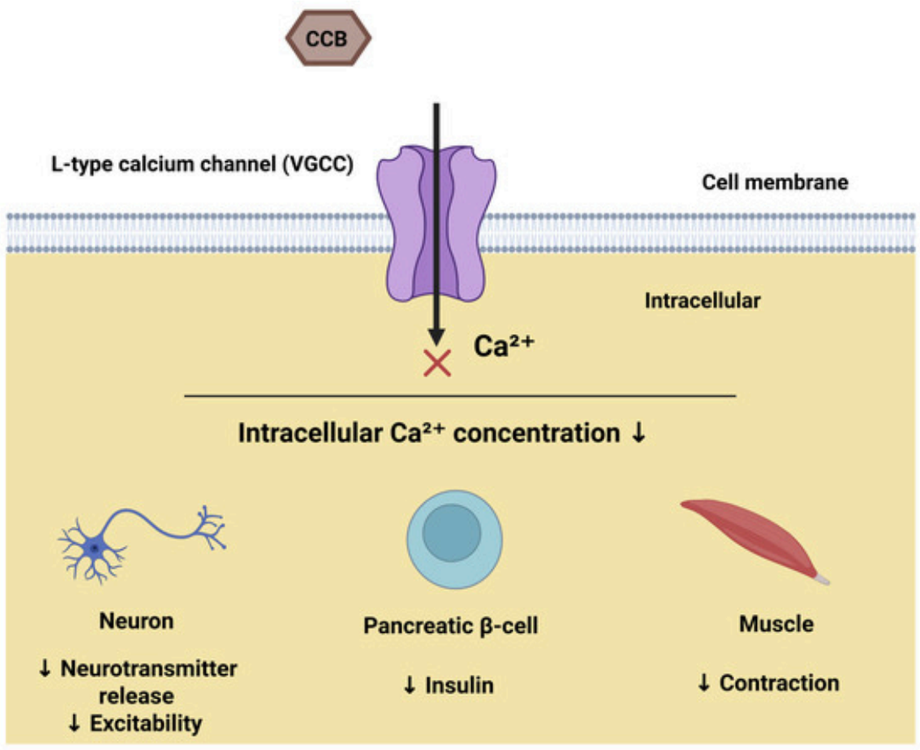


Manidipine is a third-generation, long-acting dihydropyridine **calcium channel blocker (CCB)**.

It lowers blood pressure by inhibiting calcium channels, which results in peripheral vasodilation

2 | MECHANISM OF ACTION

Manidipine is a lowers blood pressure by inhibiting calcium influx into vascular smooth muscle cells. It acts primarily by binding to L-type and T-type calcium channels, **causing vasodilation and reducing peripheral resistance**.



3 | PHARMACO KINETICS

- Manidipine is undergoes a high degree of plasma protein binding (99%) and is widely distributed to the tissues
- Oral manidipine undergoes extensive first-pass hepatic metabolism, with 63% of the drug eliminated in the feces and 31 % in the urine
- Terminal elimination half-lives after a single oral dose of manidipine 5, 10, 20 mg ranged from 3.9 to 7.95 hours. Elimination is significantly delayed in patients with severe hepatic impairment.

4 | PHARMCODYNAMICS

Therapeutic dosages of manidipine in hypertensive patients did not significantly affect norepinephrine levels, suggesting a lack of sympathetic activation. Manidipine had beneficial effects on renal function in hypertensive patients, including those with coexisting renal impairment and/or type 2 diabetes.

5 | DOSAGE

Dosing: Adult

- *Hypertension* → Oral 5-10 mg Once daily after Breakfast, titrate every 2-4 week base on patient response (Maximum 20mg/day)

6 | DOSAGE ADJUSTMENT/ CONTRAINDICATION

Dosage in renal failure: *No* dosage adjustment is required
Dosage in hepatic insufficiency: Mild hepatic dysfunction should not exceed 10 mg daily.
Pregnancy and Breastfeeding: Contraindicated in pregnancy and lactation